

Course Code : ECT 351

UVXV/MW – 24 / 1585

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

ELECTROMAGNETIC WAVES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated.
 - (3) Illustrate your answers with neat sketches.
 - (4) Assume suitable data wherever necessary.
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1.
 - (a) Recall Reflection Coefficient and VSWR, also derive the relation between Reflection Coefficient and VSWR on a loss less transmission line. 5(CO1)
 - (b) An impedance of $75 + j30 \Omega$ is connected to 100Ω line. Find the complex reflection coefficient at the load, the complex reflection coefficient and impedance at a distance of 0.25λ from the load. What is the VSWR measured on the line ? 5(CO1)
 2.
 - (a) How does phase constant vary with frequency for a uniform plane wave ? 5(CO2)
 - (b) A material has dielectric constant 25 and conductivity 2×10^6 mho/m. What is the frequency above which the material cannot behave like a good conductor ? If a Plane Wave of 10 MHz is incident on a material, effectively up to what depth can the wave penetrate the material and what will be the wavelength of the wave inside the material ? 5(CO2)
 3.
 - (a) Explain the behavior of plain waves at dielectric interface. 5(CO4)
 - (b) A light beam is incident from air to a medium with a dielectric constant 5 and relative permeability 200. If the angle of incidence is 50° , find the angle of reflection and angle of refraction. 5(CO4)

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Contd.

4. (a) Prove that attenuation factor in parallel plane wave guide is $\alpha = \text{Power lost per unit length} / (2 \times \text{Power Transmitted})$. 4(CO5)
- (b) An EM wave is propagating between parallel conducting planes in z direction, separated by 9.5 cm. Find the cut off frequency of TE_{10} , TE_{20} and TE_{30} modes. Assume air as dielectric. 6(CO5)
5. (a) The cross-section of a rectangular waveguide is 20 cm $\times$ 5 cm. Estimate the lowest order modes which will propagate on the waveguide and their cut-off frequencies. 4(CO5)
- (b) A rectangular waveguide measuring 3 cm $\times$ 4.5 cm internally has 9 GHz signal propagating in it. Calculate cut off wavelength, guide wavelength, phase velocity, group velocity and characteristic wave impedance for $TE_{1,0}$ mode. 6(CO5)
6. (a) Explain the following terms related to antenna :
- (i) Radiation Pattern.
 - (ii) Null Power Beam width.
 - (iii) Half Power Beam width.
 - (iv) Directivity.
 - (v) Gain. 5(CO3)
- (b) Transmitting and receiving antennas operating at 2 GHz with gains of 25 dB and 20 dB respectively are separated by a distance of 2 km. Assume that both antennas are identical in polarization. Estimate the received power if the transmitted power is 150 W. 5(CO3)

